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## FLINK PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Data Engineering

TOTAL: 10 QUESTIONS

**Q1** What is Flink and what problem in Data Engineering does it solve?

Threat Model

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**Q6** How does Flink handle reliability and high availability?

Observability

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**Q2** What are the core components or concepts in Flink?

Architecture

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**Q7** What observability does Flink provide out of the box?

Lifecycle

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**Q3** How does Flink compare to its main alternatives?

Configuration

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**Q8** What are common performance bottlenecks in Flink?

Compliance

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**Q4** How do you get started with Flink for a new project?

Hardening

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**Q9** What security best practices apply when running Flink?

Testing

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**Q5** What configuration options most affect Flink's behavior in production?

SSO / IdP

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**Q10** What mistakes do teams commonly make when adopting Flink?

Tuning

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# ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

## A1 Flink is a tool or platform

Mitigation

Flink is a tool or platform that automates or simplifies a specific concern in the Data Engineering domain, reducing manual effort and operational complexity for engineering teams.

## A6 Flink provides HA through leader election,

Observability

Flink provides HA through leader election, replication, or stateless horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

## A2 Flink has key building blocks that

Primitives

Flink has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

## A7 Flink exposes metrics (Prometheus endpoint or

Automation

Flink exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

## A3 Flink trades off simplicity against feature

Hardening

Flink trades off simplicity against feature richness differently than its alternatives. Choose Flink when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

## A8 Performance bottlenecks typically stem from

Governance

Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

## A4 Install Flink via its package manager

Gotchas

Install Flink via its package manager or binary release. Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

## A9 Run Flink with the principle of

Assessment

Run Flink with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

## A5 Production-critical settings typically include

Federation

Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

## A10 Common pitfalls include skipping the

Optimization

Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.